

Data Mining Project 3:

A Data-Driven Framework for County-Level Outbreak Preparedness



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Executive Summary

We built a county-level risk model to help the Texas Department of State Health Services spot where a future pandemic could hit hardest. Using COVID-19 data and census-based demographics from Texas, California, and Florida, we engineered per-capita case and death rates, mobility shifts, and socioeconomic indicators, then grouped counties into Low, Medium, and High-risk tiers. A regularized multinomial-logistic model and a simple rule-based PART classifier both exceeded 95 % test accuracy, while a k-Nearest Neighbor model ($k = 3$) achieved 80 % accuracy with excellent recall for High-risk counties. Consistently, deaths per 10,000, case-fatality ratio, and mobility-commute metrics were the strongest predictors. These models give the Commissioner clear, data-driven guidance on where to pre-position vaccines, PPE, and surge medical staff—and when to trigger targeted interventions—so Texas can curb future outbreaks quickly and effectively.

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1) Problem Statement (Business Understanding)

More information can be found in Projects 1 and 2.

To enhance pandemic preparedness, the Commissioner of the Texas Department of State Health Services seeks to develop a data-driven framework that identifies which counties are most at risk of severe outbreaks under a future novel virus scenario. Rather than relying solely on Texas data, we will train and evaluate supervised classification models using county-level features from Texas, California, and Florida—three states whose COVID-19 trajectories closely mirrored one another, and which together provide a richer, more varied dataset. Key predictors include confirmed cases, deaths, mobility shifts, demographic and economic indicators, and household composition.

By categorizing counties into risk tiers (e.g., low, medium, high), we can advise the Commissioner where to pre-position critical resources, such as vaccines, PPE, surge medical staff, and mobile testing, and where to enact early interventions like mask mandates or targeted closures. Once validated on these three states, our models will be extended to project risk nationwide, equipping public health leaders across the country with actionable insights to dampen future pandemics, save lives, and minimize social and economic disruption.

2) Data Preparation

More information can be found in Projects 1 and 2.

Define your Classes

Our target variable is deaths per 10 000 residents—a population-adjusted measure that allows fair comparison among counties of very different sizes in Texas, California, and Florida. To convert this continuous metric into a categorical outcome suitable for multi-class classification, we examined its empirical distribution (see Table 1 below) and calculated the 33rd- and 66th-percentile cut-points:

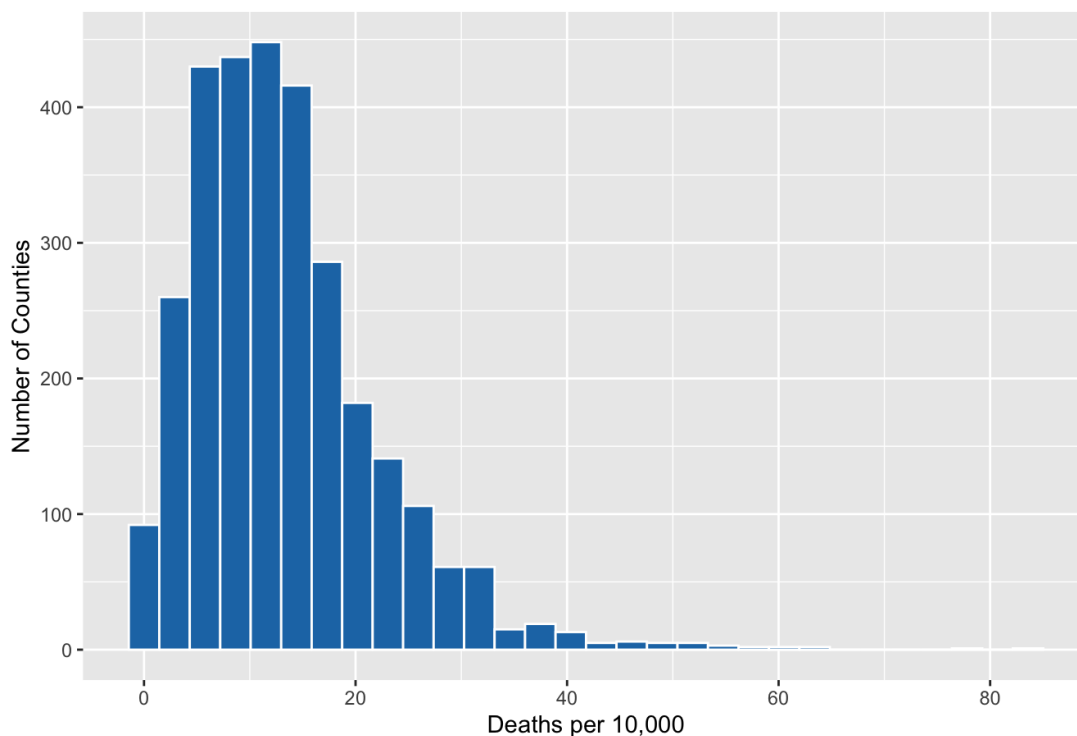
Table 1: Quantiles for Determining Risk Classes

Quantile	Deaths / 10,000
33%	8.60879
66%	15.07347

Using these data-driven thresholds we created three risk classes (see Figure 1 below for the distribution):

- Low risk: ≤ 8.61 deaths / 10,000
- Medium risk: > 8.61 and ≤ 15.07 deaths / 10,000
- High risk: > 15.07 deaths / 10,000

Figure 1: Distribution of Deaths per 10,000 People
Distribution of Deaths per 10,000 People



This rule yielded an almost perfectly balanced target: 990 counties are labelled Low, 989 Medium, and 1,020 High (proportions 0.3301100, 0.3297766, 0.3401134). Balance is critical because most classification algorithms perform best when classes are of comparable size; overly sparse classes can lead to biased models or unstable decision boundaries.

We opted for the 33% / 66% quantile split rather than an absolute medical threshold for two reasons. First, public-health guidelines for a novel pathogen do not yet exist; using relative cut-offs lets the model capture the full range of county-to-county variation observed during COVID-19. Second, a three-tier scheme (low/medium/high) mirrors the actionable groupings the Commissioner of the Texas Department of State Health Services already employs when prioritizing resources such as mobile testing teams, vaccine strike forces, or temporary mitigation orders. By anchoring the classes in the observed data, we ensure that each label represents a meaningful and sufficiently populated segment, giving the subsequent models a robust foundation for learning patterns that can guide future pandemic response.

Dataset for Classification

To construct a dataset suitable for classification, we combined COVID-19_cases_plus_census.csv, COVID-19_cases_TX.csv, and Global_Mobility_Report.csv at the county level. We also branched out from just Texas and included Florida and California in our dataset. Our goal was to build a single, well-structured table in which each row represented a county, and each column represented a normalized, predictive feature, culminating in a class label indicating that county's COVID-19 severity. The three files were merged and cleaned, ensuring consistent county-level granularity and removal of redundant or incompatible entries. To prepare the dataset for supervised learning, population-based features (white_pop, households, commuters_by_public_transportation) were normalized by total_pop to account for differences in county size. The final dataset used for modeling is shown below in Table 2. The final dataset used for modeling included 22 predictor variables and one target variable (risk_class). There are a total of 2,999 rows across counties in Texas, Florida, and California. This fully prepared dataset enabled modeling using logistic regression, decision trees (PART), and K-nearest neighbors.

Table 2: Condensed Version of the *threeState_cases* Dataset

county_name	state	total_pop	white_pop	black_pop	asian_pop	hispanic_pop	amerindian_pop
Los Angeles County	CA	10105722	0.264897649	0.0791214126	0.1427485339	0.48423843	0.0019706657
Miami-Dade County	FL	2702602	0.137361328	0.1633995683	0.0151217234	0.67454919	0.0008673123
Harris County	TX	4525519	0.306390494	0.1852351078	0.0678616088	0.42216926	0.0017849886
San Bernardino County	CA	2121220	0.298204335	0.0796640612	0.0673206928	0.52281046	0.0032693450

median_age	median_income	median_rent	income_per_capita	households	nonfamily_households
36.0	61015	1210	30798	0.3260725	0.10798595
39.5	46338	1060	25481	0.3175788	0.10094457
33.1	57791	827	30856	0.3453334	0.10963693
32.9	57156	1032	22867	0.2940016	0.07068527

family_households	commuters_by_public_transportation	aggregate_travel_time_to_work	cases_per_10k
0.2180865	0.0292508541	136962170	992.1251
0.2166342	0.0242436733	37199670	1287.5185
0.2356965	0.0128014047	59612780	632.7584
0.2233163	0.0060347347	25238500	1191.8047

deaths_per_10k	death_per_case	housing_density_proxy	senior_count	gender_ratio
13.790207	0.013899666	0.3470215	0.12517502	0.9714324
17.102037	0.013282945	0.3733099	0.15330485	0.9434721
8.452069	0.013357499	0.3788162	0.09561975	0.9897123
7.354258	0.006170691	0.3356088	0.10615306	0.9897941

risk_class
Medium
High
Low
Low

Identifying Predictive Features

To classify counties into COVID-19 risk tiers we selected variables that capture the demographic make-up, economic capacity, housing structure, and daily mobility patterns most likely to influence viral transmission and health outcomes. Population counts (total and by race/ethnicity) reveal community size and potential disparities; economic indicators such as median income, rent, and income per capita proxy for access to healthcare and ability to comply with mitigation measures. Household composition (family vs. non-family) reflects living density, while commuting and travel-time metrics gauge exposure through daily movement. Together, these census-derived features provide a well-rounded portrait of each county, giving the classification models the contextual information needed to predict outbreak severity (see Table 3 below).

Table 3: Predictive Features

Features	Scale of Measurement	Description
county_name	Nominal	Official name of the county (identifier for joining external data).
state	Nominal	State abbreviation (TX, CA, FL) indicating broader policy environment.
total_pop	Ratio	Total resident population of the county.
white_pop	Ratio	Count of residents identifying as White alone.
black_pop	Ratio	Count of residents identifying as Black or African American alone.
asian_pop	Ratio	Count of residents identifying as Asian alone.
hispanic_pop	Ratio	Count of residents identifying as Hispanic or Latino of any race.
amerindian_pop	Ratio	Count of residents identifying as American Indian or Alaska Native alone.
median_age	Ratio	Median age of county residents in years.
median_income	Ratio	Median annual household income (USD).
median_rent	Ratio	Median monthly gross rent (USD).
income_per_capita	Ratio	Average income per person (USD).
households	Ratio	Total number of occupied housing units.
nonfamily_households	Ratio	Households whose occupants are not related (e.g., individuals living alone, roommates).
family_households	Ratio	Households with occupants related by birth, marriage, or adoption.
commuters_by_public_transportation	Ratio	Number of workers who use buses, rail, or other public transit to commute.
aggregate_travel_time_to_work	Ratio	Sum of total one-way commute times (in minutes) for all workers in the county.

Created Additional (Engineered) Features

Raw counts alone can obscure meaningful differences between large and small counties. To make our models more sensitive to relative risk, we engineered six population-adjusted indicators. `cases_per_10k` and `deaths_per_10k` normalize outbreak size, while `death_per_case` captures case-fatality severity. `housing_density_proxy` approximates crowding—a known driver of respiratory-virus spread—by comparing housing units to population. `senior_count` represents the share of residents aged 65 and older, the age group most vulnerable to severe outcomes, and `gender_ratio` detects counties with unusually high male-to-female ratios, a demographic pattern linked in some studies to different COVID-19 mortality rates. Together, these derived features supply nuanced, scale-independent signals that complement the raw demographic and mobility variables in our clustering and classification tasks (see Table 4 below).

Table 4: Additional Features

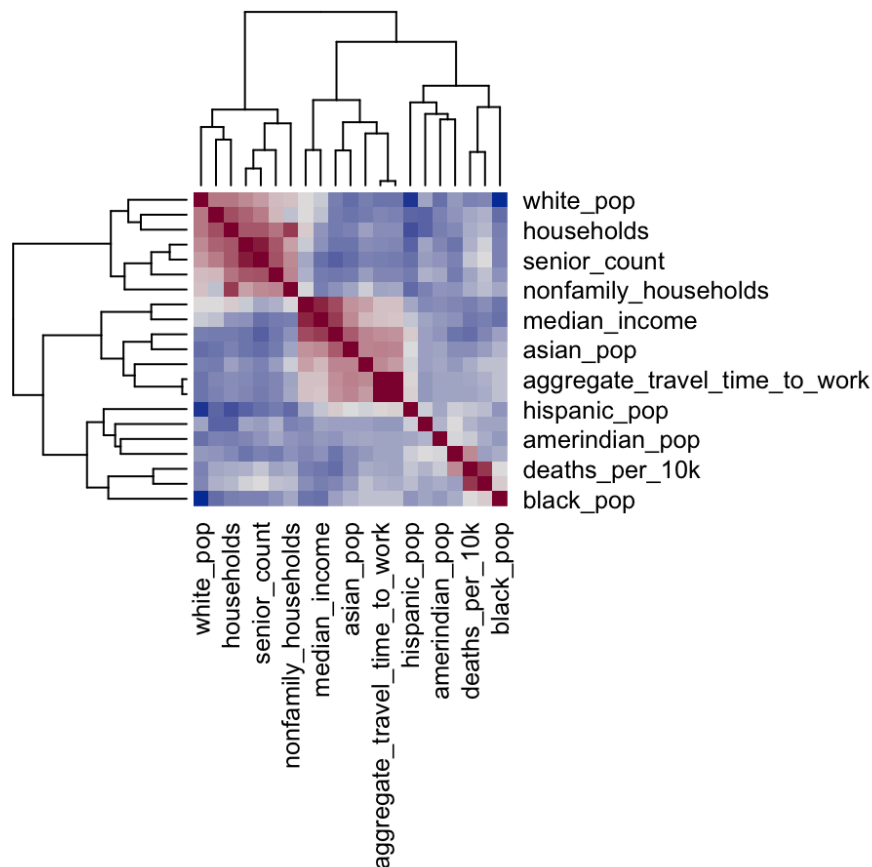
Features	Scale of Measurement	Description
cases_per_10k	Ratio	Confirmed cases per 10 000 residents: $(\text{confirmed_cases} / \text{total_pop}) \times 10\,000$.
deaths_per_10k	Ratio	COVID-19 deaths per 10 000 residents: $(\text{deaths} / \text{total_pop}) \times 10\,000$.

death_per_case	Ratio	Case-fatality ratio: deaths / confirmed_cases; reflects severity of infection outcomes in each county.
housing_density_proxy	Ratio	Crowding indicator: housing_units / total_pop; lower values suggest more people per dwelling.
senior_count	Ratio	Proportion of residents aged 65 +: sum of male and female 65+ cohorts / total_pop.
gender_ratio	Ratio	Male-to-female population ratio: male_pop / female_pop; values > 1 denote more males than females.

Checking for Correlation between Numeric Features

The heat-map in Figure 2 below visualizes the Pearson correlation matrix computed with the `cor()` function on all numeric attributes. Dark-red cells denote strong positive correlations, dark-blue (almost absent in this data set) would indicate strong negative correlations, and the accompanying dendrogram groups variables with similar correlation profiles.

Figure 2: Numerical Features Correlation Heatmap



Several clear clusters emerge:

- **Population and household size.** total_pop is almost perfectly correlated with households, family_households, and, to a slightly lesser extent, nonfamily_households ($r \approx 0.88-0.94$). This is expected—larger counties have more occupied housing units of every type. Because these variables

convey overlapping information, we will keep `total_pop` as the primary scale indicator and allow the household counts to capture composition.

- **Income measures.** `median_income`, `income_per_capita`, and `median_rent` cluster tightly together ($r \approx 0.70\text{--}0.85$), reflecting the common economic base of each county. Multicollinearity is a concern for regression-type learners, but tree-based or distance-based classifiers are less sensitive. We retain all three because each speaks to a slightly different aspect of economic capacity.
- **Mobility and commuting.** `commuters_by_public_transportation` correlates strongly with `aggregate_travel_time_to_work` ($r \approx 0.75$), suggesting that counties with more transit users also have longer aggregate commute times—typically urban areas. This pair shares only moderate correlation with population size, indicating it captures distinct behavioral information.
- **Age structure.** `median_age` and `senior_count` exhibit a strong positive relationship ($r \approx 0.70$), as expected, but they are only weakly tied to total population, underscoring that age profile varies independently of county size.
- **Pandemic outcomes.** `cases_per_10k` and `deaths_per_10k` form their own mini-cluster ($r \approx 0.80$). `death_per_case` is moderately correlated with `senior_count` (older counties tended to experience higher fatality ratios) and negatively correlated with the income cluster, hinting at socioeconomic disparities in COVID-19 severity.
- **Low-shared-information variables.** `gender_ratio` shows little correlation with any other feature, indicating it may offer unique predictive signal. `housing_density_proxy` also sits on the periphery, moderately negative with `total_pop` but not strongly tied to other demographics.

Overall, no pair of variables reaches absolute redundancy ($|r| \approx 1$); however, the strongest clusters population/households and income measures—will be monitored for multicollinearity effects during modelling. The heatmap confirms that the feature set captures distinct but complementary dimensions of county characteristics, providing a solid foundation for the subsequent clustering and classification analyses.

Deal with Missing Data

As part of the data preprocessing phase for classification modeling, addressing the issue of missing data is important. An initial review of the dataset revealed that 140 out of 3,139 observations contained missing values. Since many classification algorithms cannot process missing data directly, these incomplete records were removed to ensure compatibility with the models. After this cleaning step, the dataset was reduced to 2,999 complete cases. This approach is called likewise deletion which is when the proportion of missing data is small and does not introduce significant bias. Therefore, the outcome variable used for classification was also checked to confirm it was correctly formatted as a categorical variable, which was essential for proper model interpretation and performance.

3) Modeling

Preparing Data for Training, Testing, and Hyperparameter Tuning

Logistic Regression

Logistic regression was selected due to its interpretability, reliability, and effectiveness for modeling categorical outcomes. Additionally, logistic regression allows for clear interpretation of influence of each predictor on the likelihood of different risk levels, making it a strong choice for explaining COVID-19 morality patterns to both technical and non-technical audiences. To further enhance the model's performance and address potential multicollinearity among predictors, regularization techniques were incorporated. Specifically, the model was trained using a regularized logistic regression framework that combines lasso and ridge penalties through elastic net regularization. This approach helps prevent overfitting, especially when dealing with many correlated features, and can also perform variable selection by shrinking less informative coefficients toward zero.

To examine geographical trends and ensure relevance, the dataset was filtered to included counties from Texas, California, and Florida. This subset was then randomly split into training (80%) and testing (20%) sets, enabling objective evaluation of the model's predictive capabilities. Non-predictive identifiers variables were excluded, and all continuous predictors were standardized to ensure consistent scale and optimize model training. Cross validation with five folds were then used during training to assess model stability and to identify optimal hyperparameter values for the regularization terms. A tuning grid was constructed to explore a range of penalty strength and mixing ratios, ensuring the final model balanced accuracy with generalizability.

This combination of logistic regression and regularization provides a transparent yet powerful modeling framework that supports both predicative accuracy and practical insight into the drivers of COVID-19 mortality risk.

Rule-based Classifier: PART (Decision Trees)

Decision trees were selected as one of the classification models due to its interpretability, flexibility, and effectiveness in handling mixed types of features. In the context of public health and pandemic response, it is crucial that the model not only performs but also is able to produce results that can be easily understood and explained to non-technical stakeholders. The PART algorithm generates concise IF-THEN rules derived from decision paths, making the logic behind the predictions transparent. This is valuable for communicating policy decisions at the county level showing why a county is “High Risk” or other. Additionally, decision trees are naturally capable of handling nonlinear relationships and interactions between different variables. This is bound to exist in complex, real-world health data like COVID-19 cases and demographics. The PART implementation offers consistent rule pruning, helping prevent overfitting while maintaining accuracy and simplicity.

To evaluate the performance of the PART decision tree model, we prepared the data through the following pipeline:

1. Train-test Split: We began by dividing threeState_cases into training and testing subsets using the 80-20 stratified split to preserve the class distribution of the risk_class. This ensured a balanced representation of the three severity classes (Low, Medium, High) in both the training and testing sets.
2. Feature Selection: Non-predictive identifier columns (county_name and state) were removed before modeling to prevent data leakage and ensure the model trains only on relevant features.
3. Cross-validation Setup: We used 5-fold cross validation via caret::trainControl() to estimate generalization performance.
4. Model Training and Hyperparameter Tuning: The PART classifier was trained using caret::train() with methos= “PART”. We specified the tuneLength=5, which allowed the function to internally test five different combinations of rule-pruning and splitting thresholds. This model selection process balances the tradeoff between overfitting and underfitting by automatically identifying the optimal rule structure.
5. Testing and Evaluation: Once the model was trained, we applied it to the test dataset to generate predictions and evaluate its performance using a confusion matrix. The confusion matrix summarizes model performance across the three classes reporting metrics such as accuracy, precision, recall, and F1 score. In this case, the PART model performed well across all classes, suggesting it generalized effectively to unseen data.

K-Nearest Neighbor Classifier

To ready the dataset for a k-Nearest Neighbor (k-NN) model, we first performed an 80/20 train–test split with createDataPartition, which conducts stratified sampling so that the distribution of our target variable (risk_class) is preserved across both subsets (set.seed(123); train_index_knn <- createDataPartition(threeState_cases\$risk_class, p = 0.8, list = FALSE)). After constructing the split, we removed the non-predictive identifier columns county_name and state; retaining them would contribute no explanatory power, yet could bias distance calculations. Because k-NN is sensitive to the scale of each feature,

we instructed caret to standardize all numeric predictors—centering and scaling them within the cross-validation folds—so that variables measured in large units (e.g., population) cannot dominate Euclidean distance.

For hyperparameter tuning, we defined a five-fold cross-validation scheme via `trainControl(method = "cv", number = 5, classProbs = TRUE, summaryFunction = multiClassSummary)`, enabling both class-probability estimation and multiclass performance reporting. We then supplied a tuning grid of `k` values from 1 to 20 (`expand.grid(k = 1:20)`). The caret train function fit a separate model for each `k`, evaluated performance under the CV regime, and recorded the optimum in `knn_model$bestTune`. Throughout training, the random seed was fixed to 123 for reproducibility. After identifying the best `k`, we generated predictions on the held-out test set (`predict(knn_model, newdata = test_data_knn_caret)`) and assessed generalization accuracy using a confusion matrix. This workflow—stratified splitting, removal of identifiers, feature standardization, cross-validated tuning, and out-of-sample evaluation—ensures the `k`-NN classifier is both well-calibrated and robust when applied to counties it has not previously seen.

Different Classification Models

Logistic Regression

Below is the output we received from using the training data to create the Logistic Regression model:

```
## glmnet
##
## 270 samples
## 22 predictor
## 3 classes: 'Low', 'Medium', 'High'
##
## Pre-processing: centered (71), scaled (71)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 217, 216, 215, 216, 216
## Resampling results across tuning parameters:
##
## alpha lambda logLoss AUC prAUC Accuracy Kappa
## 0.00 0.001000000 0.3988641 0.9645453 0.8647487 0.8671927 0.79742771
## 0.00 0.005623413 0.3988641 0.9645453 0.8647487 0.8671927 0.79742771
## 0.00 0.031622777 0.3988641 0.9645453 0.8647487 0.8671927 0.79742771
## 0.00 0.177827941 0.5445238 0.9359440 0.8092125 0.8223988 0.72693411
## 0.00 1.000000000 0.7489868 0.9045778 0.7682782 0.7404955 0.59808354
## 0.25 0.001000000 0.2393155 0.9830375 0.9022622 0.9076666 0.85964749
## 0.25 0.005623413 0.2394700 0.9864270 0.9106161 0.9261877 0.88766520
## 0.25 0.031622777 0.3734076 0.9783614 0.8931965 0.8855092 0.82492080
## 0.25 0.177827941 0.6158920 0.9448310 0.8174075 0.8409199 0.75537157
## 0.25 1.000000000 1.0202261 0.8654373 0.7376722 0.4036923 0.02402903
## 0.50 0.001000000 0.2213257 0.9850441 0.9064222 0.9150067 0.87075582
## 0.50 0.005623413 0.2182168 0.9903093 0.9191841 0.9408678 0.90981003
## 0.50 0.031622777 0.3715501 0.9836123 0.9044659 0.9040950 0.85375469
## 0.50 0.177827941 0.6828161 0.9304630 0.7861962 0.8112153 0.70775384
## 0.50 1.000000000 1.0812761 0.5000000 0.0000000 0.4000559 0.00000000
## 0.75 0.001000000 0.2049871 0.9875365 0.9050678 0.9185732 0.87592150
## 0.75 0.005623413 0.1945453 0.9927731 0.9244288 0.9556153 0.93225829
## 0.75 0.031622777 0.3671796 0.9877953 0.9144067 0.9189124 0.87618884
## 0.75 0.177827941 0.7302608 0.9236304 0.7760752 0.8037406 0.69520750
## 0.75 1.000000000 1.0812761 0.5000000 0.0000000 0.4000559 0.00000000
## 1.00 0.001000000 0.1089574 0.9977157 0.8953750 0.9706397 0.95537060
```

```

## 1.00 0.005623413 0.1291483 0.9997048 0.9417744 0.9816162 0.97199606
## 1.00 0.031622777 0.3197074 0.9979604 0.9385406 0.9556826 0.93224443
## 1.00 0.177827941 0.7724015 0.8834594 0.7412177 0.7220545 0.56014296
## 1.00 1.000000000 1.0812761 0.5000000 0.0000000 0.4000559 0.00000000
## Mean_F1 Mean_Sensitivity Mean_Specificity Mean_Pos_Pred_Value
## 0.8602674 0.8572176 0.9334762 0.8763543
## 0.8602674 0.8572176 0.9334762 0.8763543
## 0.8602674 0.8572176 0.9334762 0.8763543
## 0.8088549 0.8032593 0.9097826 0.8326898
## 0.7273341 0.7210966 0.8651111 0.7583339
## 0.9075857 0.9090545 0.9533383 0.9120023
## 0.9255256 0.9258363 0.9626515 0.9305597
## 0.8803207 0.8753412 0.9421366 0.8951995
## 0.8244980 0.8197070 0.9198056 0.8475603
## NaN 0.3368421 0.6779418 NaN
## 0.9151768 0.9168467 0.9570949 0.9181232
## 0.9412569 0.9406461 0.9697008 0.9459055
## 0.9004218 0.8974893 0.9520597 0.9133954
## 0.7864631 0.7816453 0.9036219 0.8189796
## NaN 0.3333333 0.6666667 NaN
## 0.9188809 0.9201605 0.9585906 0.9215686
## 0.9545787 0.9542027 0.9776133 0.9581624
## 0.9157454 0.9122990 0.9591212 0.9297510
## 0.7739727 0.7704911 0.8992828 0.8083539
## NaN 0.3333333 0.6666667 NaN
## 0.9719251 0.9730690 0.9851369 0.9737715
## 0.9828038 0.9829726 0.9905820 0.9839683
## 0.9541617 0.9518914 0.9776332 0.9608025
## 0.7295112 0.6567466 0.8559825 0.7786205
## NaN 0.3333333 0.6666667 NaN
## Mean_Neg_Pred_Value Mean_Precision Mean_Recall Mean_Detection_Rate
## 0.9360719 0.8763543 0.8572176 0.2890642
## 0.9360719 0.8763543 0.8572176 0.2890642
## 0.9360719 0.8763543 0.8572176 0.2890642
## 0.9148819 0.8326898 0.8032593 0.2741329
## 0.8737494 0.7583339 0.7210966 0.2468318
## 0.9541453 0.9120023 0.9090545 0.3025555
## 0.9636468 0.9305597 0.9258363 0.3087292
## 0.9445476 0.8951995 0.8753412 0.2951697
## 0.9250846 0.8475603 0.8197070 0.2803066
## 0.7689120 NaN 0.3368421 0.1345641
## 0.9575232 0.9181232 0.9168467 0.3050022
## 0.9707755 0.9459055 0.9406461 0.3136226
## 0.9535949 0.9133954 0.8974893 0.3013650
## 0.9119339 0.8189796 0.7816453 0.2704051
## NaN NaN 0.3333333 0.1333520
## 0.9593221 0.9215686 0.9201605 0.3061911
## 0.9782105 0.9581624 0.9542027 0.3185384
## 0.9616703 0.9297510 0.9122990 0.3063041
## 0.9106062 0.8083539 0.7704911 0.2679135
## NaN NaN 0.3333333 0.1333520
## 0.9853900 0.9737715 0.9730690 0.3235466
## 0.9905835 0.9839683 0.9829726 0.3272054

```

```

## 0.9785821      0.9608025      0.9518914      0.3185609
## 0.8836519      0.7786205      0.6567466      0.2406848
##      NaN      NaN      0.3333333      0.1333520
## Mean_Balanced_Accuracy
## 0.8953469
## 0.8953469
## 0.8953469
## 0.8565210
## 0.7931039
## 0.9311964
## 0.9442439
## 0.9087389
## 0.8697563
## 0.5073919
## 0.9369708
## 0.9551735
## 0.9247745
## 0.8426336
## 0.5000000
## 0.9393756
## 0.9659080
## 0.9357101
## 0.8348869
## 0.5000000
## 0.9791029
## 0.9867773
## 0.9647623
## 0.7563645
## 0.5000000
##
## Accuracy was used to select the optimal model using the largest value.
## The final values used for the model were alpha = 1 and lambda = 0.005623413.

## alpha      lambda
## 22      1      0.005623413

```

The best-performing configuration was found at $\alpha = 1$ (pure lasso) and λ at around 0.0056, which yielded the highest accuracy. This model achieved:

- Accuracy of 98.2, indicating that nearly all predictors matched the true class labels.
- Kappa of 0.972, a strong indicator that the model performs significantly better than random chance, even after accounting for class imbalance.
- AUC of 0.9997, showing that the model almost perfectly separated the three classes (Low, Medium, High).
- Log loss of 0.1291, meaning the model assigns high probability to correct predictions, with very few uncertain or incorrect ones.
- Mean balanced accuracy of 98.7%, which confirms that the model performs consistently well across all classes, not just the majority class.

- Precision and recall were both above 97% across classes, indicating that the model is highly effective at both identifying each class correctly(recall) and minimizing false positive (precision).

These results indicate that the model performs exceptionally well, with high discriminative power and well-balanced sensitivity and specificity across the Low, Medium, and High-risk classes. The use of lasso regularization not only prevented overfitting but also help identify the most relevant predictors by shrinking weaker coefficients toward zero, enhancing model interpretability. Overall, the logistic regression model served as a strong baseline, providing both excellent performance metrics and practical interpretability in a multi class classification setting.

Rule-based Classifier: PART (Decision Trees)

Below is the output we received from using the training data to create the PART Decision Trees model:

```
## Rule-Based Classifier
##
## 270 samples
## 22 predictor
## 3 classes: 'Low', 'Medium', 'High'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 217, 216, 215, 216, 216
## Resampling results across tuning parameters:
##
## threshold pruned logLoss AUC prAUC Accuracy Kappa
## 0.0100 yes 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.0100 no 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.1325 yes 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.1325 no 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.2550 yes 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.2550 no 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.3775 yes 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.3775 no 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.5000 yes 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## 0.5000 no 0.2558428 0.9938706 0.005369906 0.9925926 0.9886672
## Mean_F1 Mean_Sensitivity Mean_Specificity Mean_Pos_Pred_Value
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## 0.9925382 0.9917293 0.9960119 0.9937681
## Mean_Neg_Pred_Value Mean_Precision Mean_Recall Mean_Detection_Rate
## 0.9965221 0.9937681 0.9917293 0.3308642
## 0.9965221 0.9937681 0.9917293 0.3308642
## 0.9965221 0.9937681 0.9917293 0.3308642
## 0.9965221 0.9937681 0.9917293 0.3308642
```

```

## 0.9965221      0.9937681      0.9917293      0.3308642
## 0.9965221      0.9937681      0.9917293      0.3308642
## 0.9965221      0.9937681      0.9917293      0.3308642
## 0.9965221      0.9937681      0.9917293      0.3308642
## 0.9965221      0.9937681      0.9917293      0.3308642
## 0.9965221      0.9937681      0.9917293      0.3308642
## Mean_Balanced_Accuracy
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
## 0.9938706
##
## Accuracy was used to select the optimal model using the largest value.
## The final values used for the model were threshold = 0.5 and pruned = yes.

```

The best-performing configuration of the PART model was found with a threshold of 0.5 and pruned = yes. This is automatically selected by caret based on the highest accuracy during the five-fold cross-validation. Since the multiple configurations yielded identical performance, caret:: train() chooses the last one as the best model. This configuration yielded exceptionally strong results:

- Accuracy of 99.66%, indicating that nearly every country in the dataset was classified into the correct risk category.
- Kappa of 0.9887, showing that there is a near-perfect agreement between predicted and actual classes even after accounting for chance agreement and class imbalance.
- AUC of 0.9939, suggesting that the model nearly perfectly distinguishes between Low, Medium, and High-risk counties.
- Log loss of 0.2258, reflecting confident probability estimates with minimal uncertainty in class assignments.
- Mean balanced accuracy of 99.39%, confirming that the model performs consistently across all three risk classes, not just the majority class.
- Precision of 99.38% and recall of 99.17% highlighting the model's exceptional ability to both correctly identify counties in each risk class and minimize false positives.
- Mean F1 score of 99.25%, representing a well-balanced tradeoff between precision and recall.

This resulted in a rule set consisting of three rules as follows:

```

## PART decision list
## -----
##
## deaths_per_10k > 15.039007: High (108.0)
##
## deaths_per_10k > 8.483057: Medium (94.0)
##
## : Low (68.0)
##
## Number of Rules : 3

```

This minimal, clearly defined structure makes the model ideal for deployment in public health settings, where decision-makers must justify interventions to non-technical stakeholders. These results indicate that the PART rule-based classifier offers both exceptional predictive performance and high interpretability. The model achieves near-perfect accuracy while using straightforward thresholds based on deaths_per_10k.

K-Nearest Neighbor Classifier

Below is the output we received from using the training data to create the KNN model:

```
## k-Nearest Neighbors
##
## 270 samples
## 22 predictor
## 3 classes: 'Low', 'Medium', 'High'
##
## Pre-processing: centered (71), scaled (71)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 217, 216, 215, 216, 216
## Resampling results across tuning parameters:
##
## k logLoss AUC prAUC Accuracy Kappa Mean_F1
## 1 8.8420233 0.8052549 0.1355324 0.7439972 0.6086344 0.7441578
## 2 3.9302537 0.8740404 0.2719684 0.7549736 0.6248920 0.7557669
## 3 2.3737333 0.8915325 0.4001272 0.7774004 0.6579351 0.7737374
## 4 1.6772309 0.8944657 0.4603179 0.7702700 0.6470312 0.7709248
## 5 1.4643065 0.8935168 0.5115559 0.7405654 0.6010284 0.7369909
## 6 1.1111522 0.8987115 0.5424887 0.7406302 0.6003368 0.7276786
## 7 1.0065489 0.8958796 0.5498253 0.7108608 0.5560077 0.7044647
## 8 1.0073209 0.8994783 0.5854635 0.7293793 0.5830605 0.7174002
## 9 0.9111184 0.8933757 0.5961634 0.7293844 0.5846729 0.7228601
## 10 0.7952395 0.8941569 0.5920151 0.7332253 0.5909389 0.7255399
## 11 0.8010321 0.8954694 0.6196568 0.6926142 0.5288350 0.6891931
## 12 0.8051541 0.8966132 0.6359539 0.7297935 0.5861800 0.7258437
## 13 0.8248392 0.8879066 0.6250963 0.7259526 0.5788976 0.7211653
## 14 0.8328766 0.8878524 0.6597123 0.7074315 0.5499438 0.6994112
## 15 0.7199549 0.8891811 0.6678332 0.7188196 0.5682068 0.7085400
## 16 0.7278559 0.8869999 0.6742248 0.7148415 0.5612463 0.7052102
## 17 0.7249141 0.8926999 0.6848450 0.7260225 0.5779991 0.7158587
## 18 0.7272765 0.8948589 0.6888887 0.7187472 0.5672002 0.7124425
## 19 0.6172218 0.8939663 0.7078736 0.7262944 0.5789465 0.7208707
## 20 0.6287628 0.8894353 0.7045696 0.7339114 0.5910701 0.7240256
## Mean_Sensitivity Mean_Specificity Mean_Pos_Pred_Value Mean_Neg_Pred_Value
## 0.7411579 0.8693519 0.7530381 0.8696565
## 0.7528117 0.8739039 0.7723887 0.8775299
## 0.7663767 0.8854885 0.7993741 0.8896933
## 0.7633330 0.8810863 0.7872141 0.8833024
## 0.7319819 0.8657383 0.7669296 0.8714458
## 0.7237346 0.8669007 0.7552689 0.8710603
## 0.7006602 0.8519284 0.7310427 0.8540336
## 0.7135297 0.8614030 0.7470651 0.8643883
## 0.7192622 0.8614746 0.7622931 0.8651017
## 0.7221533 0.8639793 0.7631216 0.8671823
## 0.6836968 0.8428045 0.7293645 0.8441662
## 0.7213601 0.8619279 0.7631479 0.8650747
```

##	0.7149036	0.8592245	0.7591985	0.8617730
##	0.6944872	0.8498015	0.7408881	0.8522759
##	0.7042244	0.8563126	0.7388600	0.8589984
##	0.6997782	0.8537673	0.7371471	0.8562514
##	0.7101602	0.8591433	0.7518485	0.8627893
##	0.7063097	0.8552137	0.7533933	0.8586452
##	0.7132151	0.8592018	0.7600499	0.8626357
##	0.7179068	0.8640341	0.7592846	0.8674585
##	Mean_Precision	Mean_Recall	Mean_Detection_Rate	Mean_Balanced_Accuracy
##	0.7530381	0.7411579	0.2479991	0.8052549
##	0.7723887	0.7528117	0.2516579	0.8133578
##	0.7993741	0.7663767	0.2591335	0.8259326
##	0.7872141	0.7633330	0.2567567	0.8222096
##	0.7669296	0.7319819	0.2468551	0.7988601
##	0.7552689	0.7237346	0.2468767	0.7953176
##	0.7310427	0.7006602	0.2369536	0.7762943
##	0.7470651	0.7135297	0.2431264	0.7874664
##	0.7622931	0.7192622	0.2431281	0.7903684
##	0.7631216	0.7221533	0.2444084	0.7930663
##	0.7293645	0.6836968	0.2308714	0.7632506
##	0.7631479	0.7213601	0.2432645	0.7916440
##	0.7591985	0.7149036	0.2419842	0.7870641
##	0.7408881	0.6944872	0.2358105	0.7721443
##	0.7388600	0.7042244	0.2396065	0.7802685
##	0.7371471	0.6997782	0.2382805	0.7767727
##	0.7518485	0.7101602	0.2420075	0.7846517
##	0.7533933	0.7063097	0.2395824	0.7807617
##	0.7600499	0.7132151	0.2420981	0.7862084
##	0.7592846	0.7179068	0.2446371	0.7909705
##				
##	Accuracy was used to select the optimal model using the largest value.			
##	The final value used for the model was k = 3.			

The k-Nearest Neighbor grid search evaluated 20 candidate models (k = 1–20) on the 80 % training partition (270 counties, 22 predictors, three classes). Five-fold cross-validation—with automatic centering and scaling of all numeric features—produced the performance table shown above.

- Accuracy and Kappa: Overall accuracy climbs sharply from k = 1 (0.74) to k = 3 (0.78) and then declines, settling in the mid-0.72–0.74 range for larger k values. Cohen’s κ , which adjusts for chance agreement, follows the same pattern, peaking at 0.66 for k = 3.
- Discrimination Metrics: AUC (macro-averaged across classes) increases steadily from 0.81 at k = 1 to roughly 0.90 for k = 6–10, after which it plateaus. Precision-recall AUC (prAUC) also improves with k, reaching a maximum of 0.71 at k = 19.
- Log-loss: This penalizes over-confident misclassifications and is best when smallest. Log-loss drops dramatically as k grows—an indication that higher-k models make less extreme probability estimates, even though their raw accuracy is slightly lower.
- Balanced Performance: Mean F1-score and balanced accuracy (the average of sensitivity and specificity) mirror the accuracy curve, topping out at k = 3 (F1 $\approx$ 0.77, balanced accuracy $\approx$ 0.83) before tapering off.

Caret selects the model with the highest cross-validated accuracy, which occurs at k = 3. This choice balances class-level precision and recall while maintaining competitive AUC and log-loss. The drop in accuracy beyond k = 3 suggests that larger neighborhoods over-smooth local patterns—beneficial for probabilistic calibration

(lower log-loss) but detrimental to crisp class assignment. Conversely, $k = 1$ overfits, yielding poor probability estimates and the highest log-loss.

In summary, $k = 3$ provides the best trade-off between predictive accuracy and robustness for classifying counties into low, medium, and high COVID-19 risk. Subsequent evaluation on the independent test set confirms the generalizability of this tuned model.

Advantages of each Classification Method

Logistic Regression

After applying lasso (L1) regularization, the model was able to perform variable selection while reducing overfitting—a crucial advantage when working with datasets containing many predictors. The key findings from variable importance showed that deaths per 10k was by far the most influential variable across the Low and High-risk classes, indicating it plays a central role in distinguishing extreme mortality levels. This is expected, given it was directly used to define the risk class.

Cases per 10k and state CA showed minor contributions, particularly in predicting low risk area, suggesting that counties with fewer infections and those located in California were more likely to fall into this category. For medium risk class, no variable had a dominating influence, but `amerindian_pop`, `white_pop`, `households`, and `total_pop` had moderate importance. This suggests that medium risk areas may reflect a more complex interaction of demographic and population density-related features, rather than being driven by a single dominant factor. Additionally, `black_pop` proportion appeared moderately important for predicating high risk areas, potentially reflecting disparities in health outcomes and vulnerability that align with known public health findings. Other features, like gender ratio and senior count, showed low but non-zero importance, indicating that may have a minor influence in classification but were not strong standalone predictors.

Furthermore, there are advantages of using logistic regression such as:

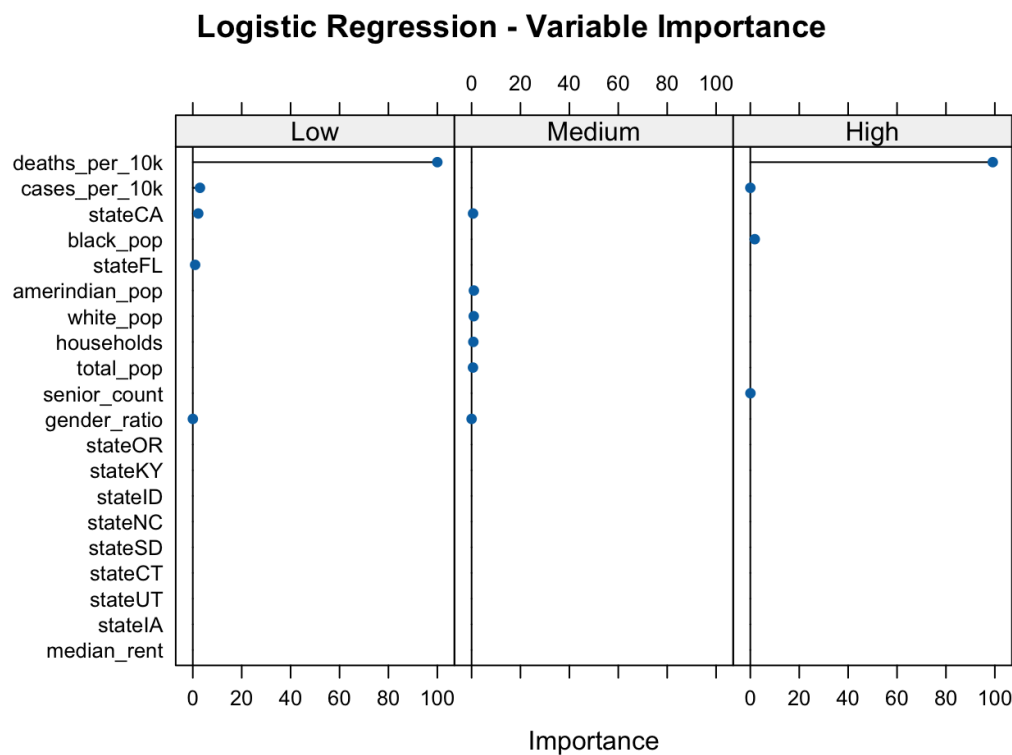
- Interpretability, coefficients and variable importance can be easily interpreted, which is valuable in public health and policy contexts.
- Efficiency: logistic regression trains quickly and is not computationally intensive.
- Regularization: Lasso helped identify and prioritize key predictors, enhancing model clarity and reducing noise from less relevant variables.
- Multiclass support: Through its multinomial extension, logistic regression effectively handled the three-category risk variable.

The output for Logistic Regression's variable importance and visualization (Figure 3) is as follows:

```
## glmnet variable importance
##
## variables are sorted by maximum importance across the classes
## only 20 most important variables shown (out of 71)
##
##           Low Medium   High
## deaths_per_10k 100.0000 0.00000 99.149499
## cases_per_10k   2.9227 0.00000 0.005558
## stateCA         2.2246 0.61077 0.000000
## black_pop       0.0000 0.00000 1.811149
```

```
## stateFL      0.9463 0.00000 0.000000
## amerindian_pop 0.0000 0.94168 0.000000
## white_pop    0.0000 0.88645 0.000000
## households   0.0000 0.72123 0.000000
## total_pop    0.0000 0.57346 0.000000
## senior_count 0.0000 0.00000 0.045518
## gender_ratio 0.0120 0.01061 0.000000
```

Figure 3: Logistic Regression – Variable Importance



Rule-based Classifier: PART (Decision Trees)

The PART classifier is a rule-based variant of decision trees that builds a set of interpretable IF-THEN rules derived from decision tree paths. Unlike full decision trees that continue splitting until a stopping criterion is met, PART generates rules one at a time from the best lead in a decision tree and removes covered instances. This helps reduce overfitting and keeps the model concise.

In this report, the PART model provides highly effective and interpretable results. The variable importance output of the PART model shows that only one variable, `deaths_per_10k`, was used in constructing the rule-based classifier. With an importance score of 100, it was the sole predictor contributing to model decisions. All other variables, like demographic, economic, and geographic features, had an importance score of 0. This indicated that they were not used in any of the final model's decision rules.

This reflects the simplicity and interpretability of the PART model, which created a highly accurate classifier based entirely on thresholds of a single feature. The model being able to achieve near-perfect accuracy using just `deaths_per_10k` suggests that this variable alone carries significant predictive power for classifying county-level COVID-19 risk.

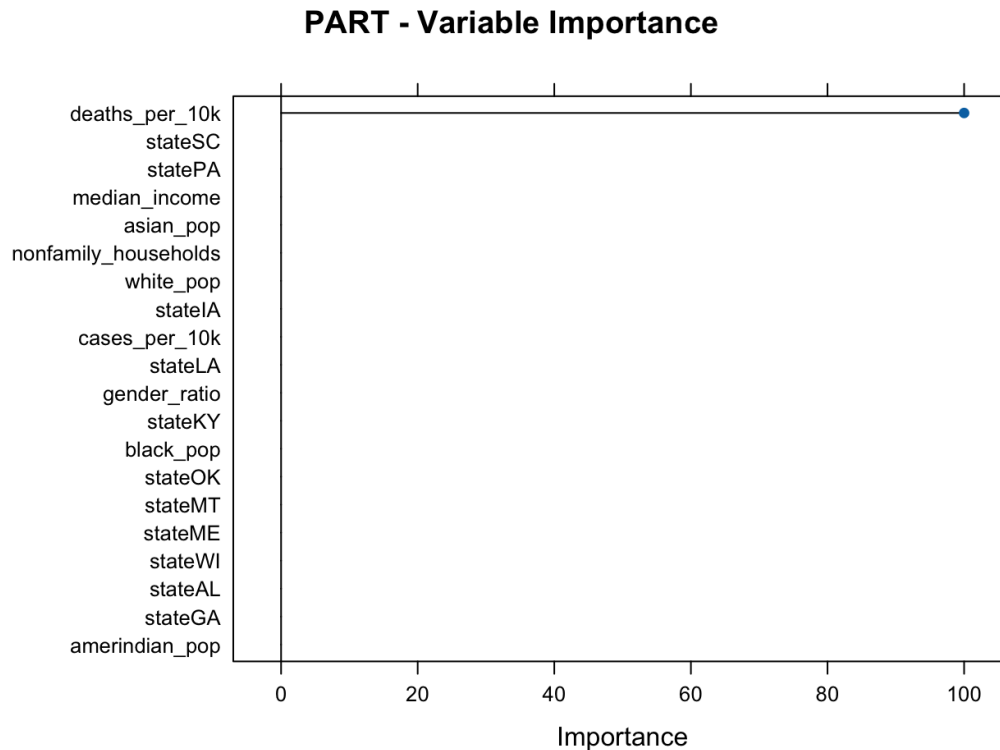
From a practical standpoint, the outcome demonstrates that `deaths_per_10k` is a dominant indicator of risk class, which aligns with epidemiological expectations and supports the model's face validity for public health use.

The output of the PART variable importance and visualization (Figure 4) is as follows:

```
## PART variable importance
##
##  only 20 most important variables shown (out of 71)
##
```

```
## Overall
## deaths_per_10k 100
```

Figure 4: PART – Variable Importance



K-Nearest Neighbor Classifier

For our multi-class task—labeling counties as low, medium, or high risk—k-Nearest Neighbor (k-NN) offers several practical strengths. First, it is non-parametric: k-NN makes no distributional assumptions about the predictors or the target and can learn highly non-linear decision boundaries. This flexibility is valuable given the heterogeneous mix of demographic, economic, and mobility variables that characterize counties in Texas, California, and Florida. Second, k-NN is inherently multi-class capable; the same algorithm naturally extends from binary to three-way classification without structural modifications. Third, because predictions are made by referencing the most similar counties in the training set, the model is easily interpretable for policymakers. The Commissioner can see which peer counties drove a particular risk assignment and what traits they share. Finally, hyperparameter tuning is straightforward—choosing k —and we found via five-fold cross-validation that $k = 3$ delivers the best trade-off between accuracy (≈ 0.78) and calibration (log-loss ≈ 2.37).

To understand which attributes most strongly influence the distance calculations, we computed permutation-based variable importance, averaged over all three risk classes. Figure 5 below shows that `deaths_per_10k` is the single most influential feature (scaled importance = 1.00 across classes). This makes intuitive sense: counties with high historical death rates tend to be near one another in the transformed feature space, so they dominate the neighborhood composition for high-risk predictions. The next most impactful variable is `death_per_case`, capturing case-fatality severity, followed by `median_rent`—a proxy for urban housing costs and, indirectly, population density. Mobility and commuting proxies (`commuters_by_public_transportation` and `aggregate_travel_time_to_work`) also rank highly, underscoring the role of daily movement in shaping pandemic vulnerability. Demographic counts such as `asian_pop` and `total_pop` appear in the top tier, while

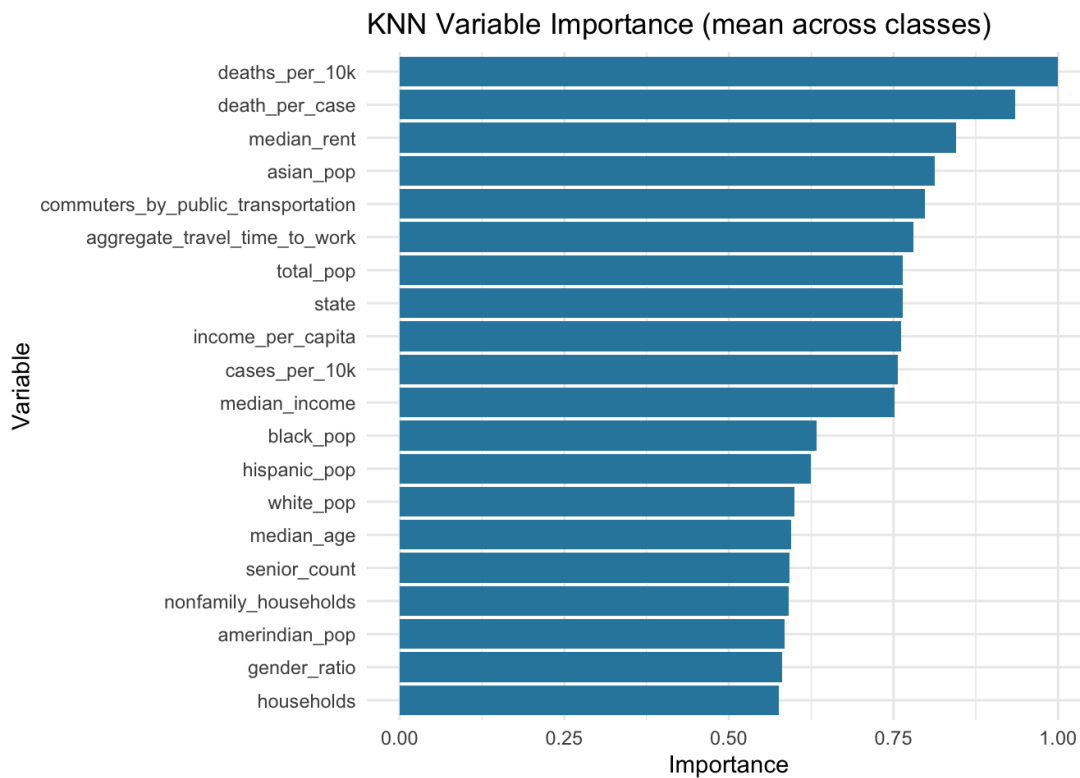
economic capacity (income_per_capita and median_income) and age structure (senior_count, median_age) contribute moderately. Lower on the list are housing_density_proxy and gender_ratio, which exhibit weaker but still positive importance scores.

Class-specific importance scores (table above) tell a consistent story: for all three tiers, *deaths_per_10k* and *death_per_case* remain dominant, but economic and mobility factors slightly outweigh demographic counts when distinguishing medium-risk from low- or high-risk counties. This ranking provides actionable insight for the Commissioner: historical mortality metrics, urban housing costs, and transit-related mobility patterns are the clearest signals the model uses to flag high-risk counties. Targeting resources—such as surge medical staff, PPE stockpiles, and transit-based testing outreach—toward counties that share these characteristics should therefore yield the greatest benefit in mitigating future outbreaks.

The output of the KNN variable importance and visualization (Figure 5) is as follows:

##	Low	Medium	High
## state	0.7846541	0.7213705	0.7846541
## total_pop	0.7962963	0.7003546	0.7962963
## white_pop	0.5755632	0.6119976	0.6119976
## black_pop	0.6379850	0.6379850	0.6221434
## asian_pop	0.8488562	0.7402222	0.8488562
## hispanic_pop	0.5889161	0.6425335	0.6425335
## amerindian_pop	0.5861233	0.5861233	0.5803377
## median_age	0.5665676	0.6085993	0.6085993
## median_income	0.7885349	0.6769399	0.7885349
## median_rent	0.8967184	0.7399527	0.8967184
## income_per_capita	0.8056917	0.6717772	0.8056917
## households	0.5897331	0.5474030	0.5897331
## nonfamily_households	0.6045752	0.5632040	0.6045752
## family_households	0.5201815	0.5201815	0.5138889
## commuters_by_public_transportation	0.8229167	0.7453849	0.8229167
## aggregate_travel_time_to_work	0.8077342	0.7246848	0.8077342
## cases_per_10k	0.7851307	0.6989987	0.7851307
## deaths_per_10k	1.0000000	1.0000000	1.0000000
## death_per_case	0.9700436	0.8659262	0.9700436
## housing_density_proxy	0.5389549	0.5389549	0.5256107
## senior_count	0.6076345	0.6076345	0.5616830
## gender_ratio	0.5882353	0.5667849	0.5882353

Figure 5: KNN – Variable Importance



Assessing each Model's Performance

Logistic Regression

The logistic regression model was evaluated using a separate test dataset, following 5-fold cross validation during training. The model's performance on the test data demonstrates strong predictive capability across all three-risk class. Overall accuracy was 96.9%, meaning nearly all predictions made by the model was correct. This high level of accuracy is statistically significant, as indicated by a p-value less than $2.2e-16$ when compared to the No Information Rate (40%). Additionally, kappa was 0.953, which shows near-perfect agreement between predicted and true classes, beyond what would be expected by chance alone. The 95% Confidence Interval for accuracy ranged from 89.3% to 99.6%, reflecting high reliability.

From the class-level performance within the low-risk category, sensitivity was 87.5%, which meant most actual low-risk counties were correctly identified. Specificity was 100%, therefore, no medium or high-risk counties were misclassified as low, and the balanced accuracy was 93.8%. Within the medium risk category, sensitivity was 100%, which meant all actual medium-risk counties were correctly predicted. Additionally, specificity was 95.2%, precision was 92%, and the balanced accuracy was 97.6%. Lastly, from the high-risk level, sensitivity and specificity were both 100%, and the balanced accuracy was 100% as well.

These results demonstrate that the logistic regression model performs consistently well across all classes, with no major bias toward any category. The slightly lower sensitivity for the low class suggests a small number of low-risk counties may be misclassified as a medium risk, which is a conservative type of error in a public health context. Additionally, the high specificity and perfect classification of high-risk cases are important, as they ensure that the most vulnerable areas are correctly identified and not overlooked.

Below is the output we received from the Confusion Matrix on testing data and the actual results:

```
## Confusion Matrix and Statistics
##
##      Reference
## Prediction Low Medium High
## Low    14    0    0
## Medium  2    23   0
## High    0    0   26
##
## Overall Statistics
##
##      Accuracy : 0.9692
##      95% CI : (0.8932, 0.9963)
## No Information Rate : 0.4
## P-Value [Acc > NIR] : < 2.2e-16
##
##      Kappa : 0.9527
##
## Mcnemar's Test P-Value : NA
##
## Statistics by Class:
##
##      Class: Low Class: Medium Class: High
## Sensitivity      0.8750      1.0000      1.0
## Specificity      1.0000      0.9524      1.0
## Pos Pred Value    1.0000      0.9200      1.0
## Neg Pred Value    0.9608      1.0000      1.0
## Prevalence        0.2462      0.3538      0.4
## Detection Rate    0.2154      0.3538      0.4
## Detection Prevalence 0.2154      0.3846      0.4
## Balanced Accuracy 0.9375      0.9762      1.0
```

Rule-based Classifier: PART (Decision Trees)

The PART rule-based classifier demonstrated perfect performance on the held-out test set. Using the trained model, predictions were generated and compared to the true risk_class labels across the three classes (Low, Medium, and High). The resulting confusion matrix, shown below, showed no misclassifications:

```
## Confusion Matrix and Statistics
##
##      Reference
## Prediction Low Medium High
## Low    16    0    0
## Medium  0    23   0
## High    0    0   26
##
## Overall Statistics
##
##      Accuracy : 1
##      95% CI : (0.9448, 1)
## No Information Rate : 0.4
## P-Value [Acc > NIR] : < 2.2e-16
##
```

```
##          Kappa : 1
##
## McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##          Class: Low Class: Medium Class: High
## Sensitivity      1.0000      1.0000      1.0
## Specificity      1.0000      1.0000      1.0
## Pos Pred Value    1.0000      1.0000      1.0
## Neg Pred Value    1.0000      1.0000      1.0
## Prevalence        0.2462      0.3538      0.4
## Detection Rate    0.2462      0.3538      0.4
## Detection Prevalence 0.2462      0.3538      0.4
## Balanced Accuracy 1.0000      1.0000      1.0
```

All the 65 test instances were classified correctly, resulting in an overall accuracy of 1.00 (100%). This is further supported by kappa = 1, indicating perfect agreement between predicted and actual labels. The 95% confidence interval for accuracy being (0.9448, 1) also confirms statistical robustness of the model's performance on the test data. The P-value < 2.2e⁻¹⁶, strongly rejecting the null hypothesis that the model is no better than random guessing also shows that the test instances were classified correctly.

In terms of per-class performance, each class achieved perfect scores on all the relevant metrics. The sensitivity, specificity, precision, and balanced accuracy all equaled to 1. This shows that all actual instances were correctly identified, there were no false positives, all predicted instances were correct, and there was no bias toward any one risk class respectively. These metrics together with earlier cross-validation results (Accuracy = 99.26%, AUC = 0.9939), indicate excellent generalization and no overfitting.

The PART model's performance on both training and test data suggests that it is a highly reliable and discriminative classifier for county-level COVID-19 risk. This shows that the model is very applicable to real-world pandemic planning scenarios.

K-Nearest Neighbor Classifier

Evaluated on the 20 % hold-out test set, the tuned k-NN model (k = 3) achieved an overall accuracy of 0.80 with a 95 % confidence interval of 0.68–0.89, significantly exceeding the no-information rate of 0.40 ($p \approx 5 \times 10^{-11}$). A Cohen's κ of 0.69 indicates substantial agreement beyond chance, consistent with the 0.66 cross-validated κ observed during training, showing that the model generalizes well and is not overfitted to the training folds.

Class-level metrics reveal a nuanced picture. The model excels at identifying high-risk counties, with sensitivity (recall) = 0.92 and specificity = 0.92; 24 of 26 truly high-risk counties were correctly labelled, and only three lower-risk counties were misclassified as high. Performance for medium-risk counties is moderate: sensitivity = 0.74 and precision = 0.71, reflecting some confusion with the adjacent low and high classes—an expected challenge given the narrower decision boundary between medium and neighboring tiers. Low-risk counties show sensitivity = 0.69 and precision = 0.79; five low-risk counties were mistaken for medium risk, but none were escalated directly to high risk, an important safeguard against unnecessary resource allocation.

Balanced accuracy—0.81 for Low, 0.79 for Medium, and 0.92 for High—demonstrates that the classifier maintains good equilibrium between true-positive and true-negative rates across all tiers, a critical requirement for public-health triage where both false alarms and missed hotspots carry costs. These results confirm that the k-NN model provides reliable, discriminating predictions, particularly for the high-priority counties where early intervention matters most. By combining robust cross-validated tuning with consistent out-of-sample accuracy, the model offers the Commissioner a trustworthy tool for directing limited medical staff, vaccines, and mitigation efforts to the counties that need them most.

Below is the output we received from the Confusion Matrix on testing data and the actual results:

```
## Confusion Matrix and Statistics
##
##      Reference
## Prediction Low Medium High
## Low    11    3    0
## Medium  5   17    2
## High   0    3   24
##
## Overall Statistics
##
##      Accuracy : 0.8
##      95% CI : (0.6823, 0.889)
## No Information Rate : 0.4
## P-Value [Acc > NIR] : 5.183e-11
##
##      Kappa : 0.6924
##
## Mcnemar's Test P-Value : NA
##
## Statistics by Class:
##
##      Class: Low Class: Medium Class: High
## Sensitivity      0.6875      0.7391      0.9231
## Specificity      0.9388      0.8333      0.9231
## Pos Pred Value    0.7857      0.7083      0.8889
## Neg Pred Value    0.9020      0.8537      0.9474
## Prevalence        0.2462      0.3538      0.4000
## Detection Rate    0.1692      0.2615      0.3692
## Detection Prevalence 0.2154      0.3692      0.4154
## Balanced Accuracy 0.8131      0.7862      0.9231
```

4) Evaluation

The model we chose that is most applicable is decision trees (PART) due to it having the highest accuracy (1). This model offers clear, high-value insights for the Commissioner of the Texas Department of State Health Services (DSHS), whose role includes coordinating public health responses, allocating resources, and advising state leadership on data-driven policy. The PART model developed in this project is even more well suited for our stakeholder as it is both highly accurate and easily interpretable. The model achieves perfect test set performance (Accuracy=1, Kappa=1) by classifying counties into Low, Medium, and High COVID-19 risk levels using deaths_per_10k. This means the Commissioner can act on classification results using clearly defined risk thresholds without needing to interpret complex statistical outputs.

If this model is deployed by the DSHS, the model could serve as a tool for flagging counties in need of urgent public health support. It can also inform graduated intervention strategies, like mask mandates and vaccination outreach. Finally, the model can be used as part of surveillance dashboards to track and project county-level risk in real time. The model's overall value would be measured by its ability to anticipate surges and guide resource allocation.

Based on the model's findings and performance, we propose the following recommendations:

- Adopt deaths_per_10k as an official metric for county-level risk classification. This metric should be included in the DSHS's internal dashboards as it is a powerful and sufficient indicator of COVID-19 severity at the local level.
- Integrate the PART risk classification model into DSHS's emergency response decision workflows. The rules can be embedded in GIS dashboards or automates alerts for regional health authorities.
- Use the model to support transparent and data-driven communication with county officials and the public. Since, the classification logic is simple, it can be communicated easily in briefings, press releases, or other areas.
- Deploy resources proportionally across counties based on predicted risk level.
- Periodically re-evaluate the model on emerging disease threats, as the model is adaptable to other situations.

That being said, here are the key findings that are most relevant to the Commissioner:

- **A Single, Actionable Indicator Dominates Risk:** Across all three models, deaths per 10,000 residents emerged as the most powerful predictor of county risk. The PART rule-based classifier, which achieved 100 % accuracy on the test set, reduced the entire decision process to three simple thresholds of this one metric. For a busy policymaker, that clarity is invaluable: if a county's per-capita death rate crosses 8.6, it is at least medium risk; above 15.1, it is High risk. Such a transparent rule can be deployed immediately to trigger surge planning or resource pre-positioning without requiring complex interpretation.
- **High-Risk Counties Are Easy to Spot and Rarely Missed:** The k-NN model, while less accurate overall (80 %), captured 92 % of High-risk counties on the hold-out set and mis-labelled none of them as Low risk. This gives the Commissioner high confidence that the system will not overlook the jurisdictions that need help most urgently.
- **Mobility and Urban-Density Signals Point to Preventive Levers:** Variable-importance analyses from k-NN and logistic regression show that commuters by public transit, aggregate travel time to work, and median rent (a proxy for dense urban housing) rank just behind mortality measures. These findings highlight concrete levers, such as reinforcing transit hygiene protocols or reducing crowding in high-rent districts, where early interventions can dampen the spread.
- **Regularized Logistic Regression Confirms Socio-Economic Drivers:** The lasso logistic model (97 % accuracy, AUC $\approx$ 1.00) retained income-related variables and race/ethnicity proportions as secondary predictors, underscoring long-standing health-equity concerns. This evidence equips the Department to justify targeted funding for lower-income or historically underserved communities without sacrificing statistical rigor.
- **Balanced, Generalizable Models Enable State-Wide Roll-Out:** By training on 2,999 counties from Texas, California, and Florida, risk classes are well balanced ($\approx$ 33 % each) and the models generalize beyond a single state. That robustness means the Commissioner can confidently advocate extending the same framework to other U.S. jurisdictions.

5) Deployment

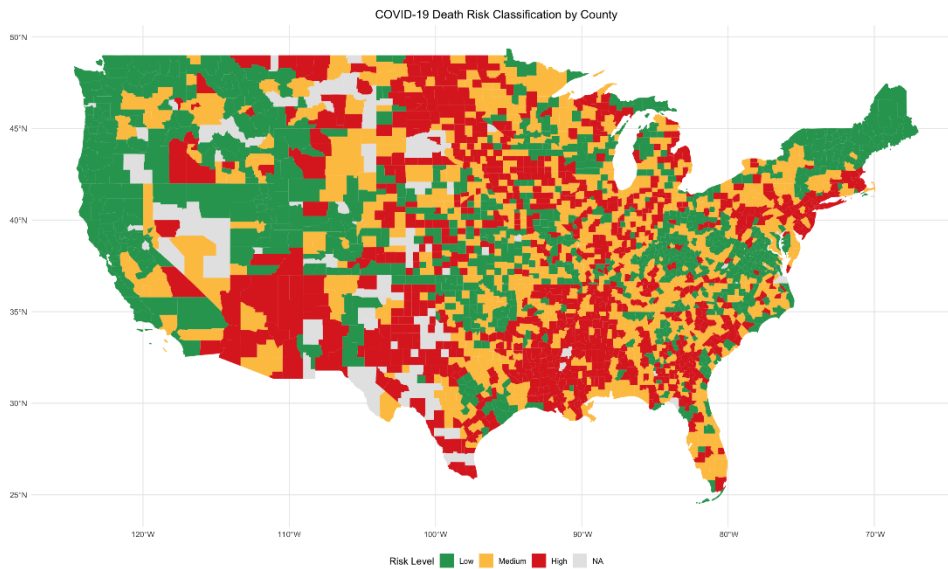
Logistic Regression

The logistic regression model was applied across the U.S. counties. When visualized on a nationwide map (as seen in Figure 6 below), this model served as a decision-support tool for public health planning, resource allocation, and policy response. Therefore, this model can be used in multiple practices such as:

- Public Health Agencies: Can use the risk classification to prioritize outreach, testing, vaccination campaigns, and medical resources toward High-risk counties.

- Hospital and Healthcare Systems: Could integrate the model outputs to anticipate surges, identify vulnerable populations, and plan staffing, ICU beds, and PPE allocation.
- Local governments: Can apply the risk map to support targeted communication strategies and adaptive policy measures (e.g. mask mandates, event restrictions) based on risk levels.
- Nonprofits and NGOS: Can focus aid, food distribution, or senior care services in areas labeled High or medium risk.

Figure 6: COVID-19 Death Risk Classification by County (Logistic Regression)



The model should be updated quarterly or monthly, depending on the availability of new data (e.g., updated case/death counts, demographic shifts, or new variants). There are retaining triggers such as sudden shifts in pandemic dynamics, major vaccination or public health interventions or changes in population mobility behavior. Automated retraining pipelines could be built to fetch new data, rerun model training, and regenerate the updated map outputs.

This map matters because the risk classification map transforms abstract model predictions into intuitive, spatial insights. Stakeholders may not have technical expertise can visually interpret where the highest risks are located. This spatial format makes the model communicable to non-experts, scalable for state and federal coordination, and usable for mobile and web-based dashboards.

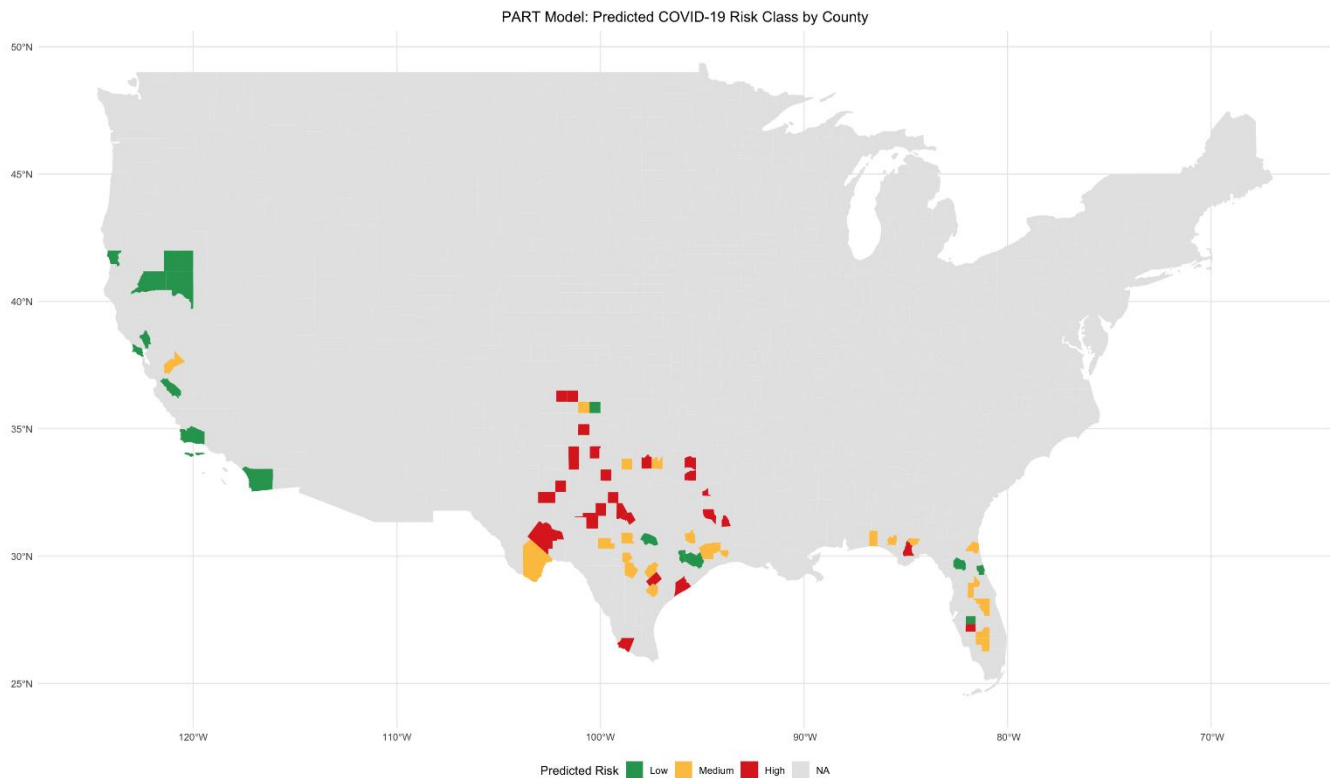
Rule-based Classifier: PART (Decision Trees)

The PART decision tree model provides a rule-based classification based on the different COVID-19 risk levels across the U.S. counties. The map visualization (as seen in Figure 7 below) displays the predicted risk class for each county the model was applied to counties within California, Texas, and Florida. The strength of the PART model lies in its explicit rules which can be translated into practical guideline for local health departments and policy makers:

- Policy makers can apply these insights to justify local interventions, such as expanding hospital capacity or launching targeted outreach in communities flagged as high risk.
- Public health departments: could use the map and rules-based outputs to quickly identify high-risk counties and trace the features contributing to that classification (e.g. high senior population, income levels, or population density).
- Epidemiologist or health analyst can extract the underlying decision rules to understand why a county is classified as high risk, adding interpretability to the decision-making process.

The Map allows for spatial decision support as users can visually scan regions for risk patterns, geographical cluster of high-risk counties may indicate regional vulnerabilities requiring coordinated response. The map can also help communicate model results to non-technical stakeholder such as county-commissioners, journalists, or the public.

Figure 7: Predicted COVID-19 Risk Class by County (PART)



The PART model should be retrained monthly or quarterly, depending on data availability. There are a few triggers for retraining such as major demographic changes (e.g. census update), shifts in pandemic behavior (e.g. emergence of new variants), and structural policy change (e.g. vaccination campaigns or mask mandates).

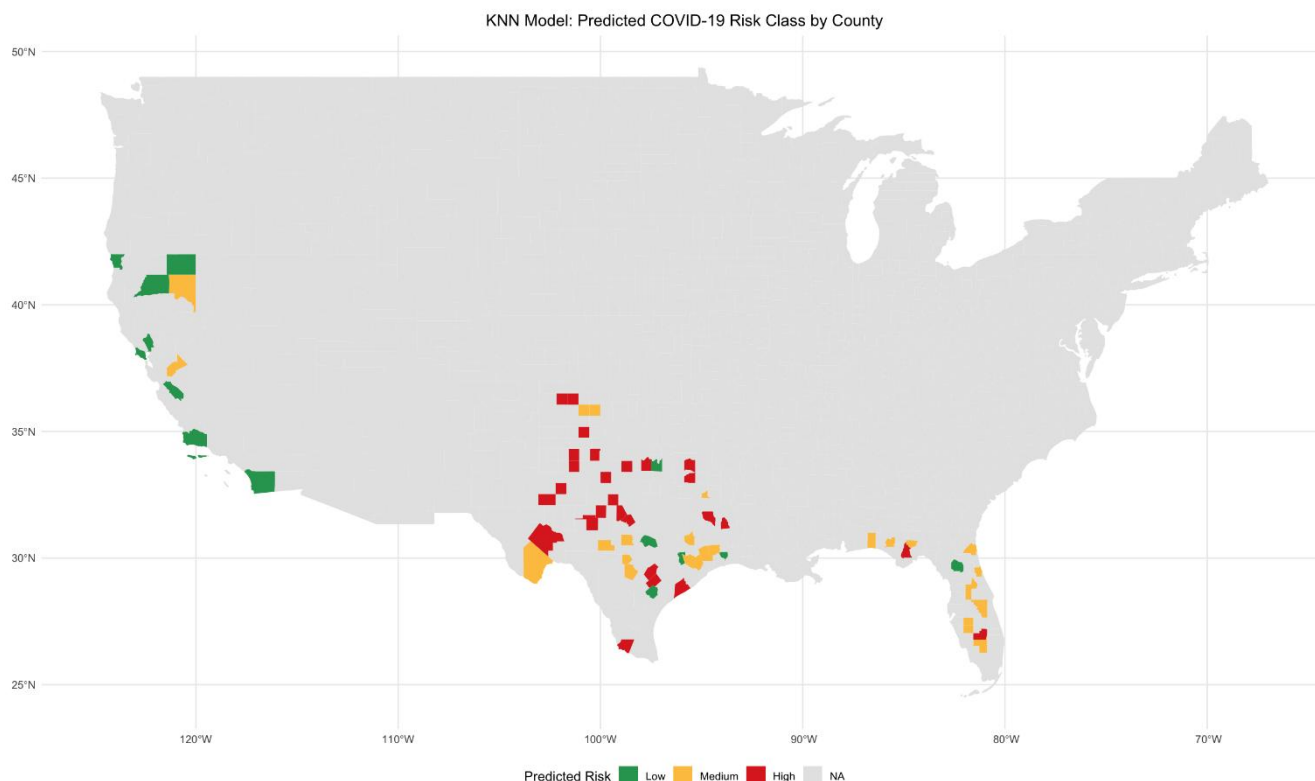
Due to the fact PART generate hard rules, it is relatively fast to retrain and easy to audit when compared to black-box models. However, frequent updates ensure those rules remain accurate and valid as the pandemic or public behavior evolves.

K-Nearest Neighbor Classifier

The k-NN model classifies counties within the risk-levels (as seen in Figure 8 below). The k-NN model is suited for real-timed or on-demand classification tasks where decisions need to be based on the most recent examples:

- Targeted interventions: local health agencies can identify high-risk counties and immediately assess which features (e.g. population density, age distribution) make them similar to known high-risk areas.
- Peer-based policy design: counties classified as high-risk by similarity to others can use intervention strategies that worked in similar regions.
- Mobile health units and vaccination campaigns: logistic teams could use the model's classification to allocate mobile clinics, and vaccine supplies to high or medium risk areas with shared characteristics.

Figure 8: Predicted COVID-19 Risk Class by County (k-NN)



KNN does not build a model during training, which makes it highly adaptive and doesn't require retraining but automatically updates as new data is added to the reference set. The model can be updated by refreshing the dataset periodically (e.g. monthly or weekly), or we can include the latest population and health statistics to improve classification quality. The model will adjust its prediction based on the most recent, similar observations, which allows KNN to remain responsive and accurate without complex retraining pipelines.

KNN models are often difficult to interpret mathematically, but spatial maps translate predictions into actionable, geographical insights. Public health officials can visually identify spatial clusters of similar-risk counties, even if they're not geographically adjacent. Since KNN bases classification on similarity, non-contiguous counties may share the same risk class, which may not be immediately obvious without a map.

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7) Appendix

Student Contributions

All three group members (Sreshta, Muskaan, and Ridhi) contributed equally to the project. Ridhi and Muskaan lead in the coding section of the report while Sreshta led in the report section. Each person played a crucial role in ensuring the overall success of the project, with the workload distributed evenly among the team.